

```
udayasri@SURABHIUDAYASRI:~$ cd OSLab
udayasri@SURABHIUDAYASRI:~/OSLab$ nano permission.c
udayasri@SURABHIUDAYASRI:~/OSLab$ gcc permission.c -o permission
udayasri@SURABHIUDAYASRI:~/OSLab$ ./permission
File created successfully.
udayasri@SURABHIUDAYASRI:~/OSLab$ ls
Makefile      fork          hello        obj           process.c    shell.c
Practical04   fork.c       hello.c      permission    sample.txt   src
bin           fork_program include       permission.c shell
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw-r--r-- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

- 1) Give Read and Write only to the owner

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 600 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw----- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Read + Write = 4 + 2 = 6

Group - Nothing = 0

Others - Nothing = 0

- 2) Give Read and Write to the owner and group

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 660 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Read + Write = 4 + 2 = 6

Group - Read + Write = 4 + 2 = 6

Others - Nothing = 0

- 3) Only owner can read and write

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 600 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw----- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Read + Write = 4 + 2 = 6

Group - Nothing = 0

Others - Nothing = 0

- 4) Owner read/write, everyone else read-only

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 644 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw-r--r-- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Read + Write = 4 + 2 = 6

Group - Read = 4

Others - Read = 4

5) Owner can do everything, others can only read and execute

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 755 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rwxr-xr-x 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ |
```

Owner → Read + Write + Execute = 4 + 2 + 1 = 7

Group - Read + Execute = 4 + 1 = 5

Others - Read + Execute = 4 + 1 = 5

6) Owner and group can read/write

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 660 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Read + Write = 4 + 2 = 6

Group - Read + Write = 4 + 2 = 6

Others - Nothing = 0

7) Only owner can execute

```
udayasri@SURABHIUDAYASRI:~/OSLab$ chmod 100 sample.txt
udayasri@SURABHIUDAYASRI:~/OSLab$ ls -l sample.txt
---x----- 1 udayasri udayasri 28 Aug 17 05:57 sample.txt
```

Owner - Execute = 1

Group - Nothing = 0

Others - Nothing = 0

8) Owner: read + write; Group: nothing; Others: nothing

Owner - 4 + 2 = 6

Group - 0

Others - 0

Answer = 600

9) Owner: read + write; Group: read; Others: read

Owner - $4 + 2 = 6$

Group - 4

Others - 4

Answer = 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute

Owner - $4+2+1=7$

Group - $4+1=5$

Others - $4+1=5$

Answer = 755

11) Owner: read + write; Group: read + write; Others: nothing

Owner - $4 + 2 = 6$

Group - $4 + 2 = 6$

Others - 0

Answer = 660

12) Everyone: read only

Owner - 4

Group - 4

Others - 4

Answer = 444